Chapter 9: Estimation and Confidence Intervals

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Introduction

Consider a dataset of the population of all 800 schools in Bahrain from which we have their math grade:

School i	**Mathematics Grade**	**Pass Indicator** X_i
School 1	87	1
School 2	58	1
School 3	46	0
School 4	48	0
School 5	80	1
School 6	67	1
School 7	35	0
School 8	67	1
School 9	82	1
	⋮	
School 800	82	1

Introduction

Since *school_data.xlsx* shows data on all schools, then this is the population dataset. Therefore the average math score and its standard deviation is:

$$\mu = 71 \qquad \sigma = 9.95$$

Since this is the data for all schools, the parameter μ represents the **true** average of the population.

Introduction

Suppose however we did not have access to this population data. Instead, we only have a sample of just 5 schools:

School i	**Average Mathematics Grade** x	**Governorate**
School 342	88	Capital
School 675	68	Muharraq
School 23	56	Muharraq
School 331	48	Southern
School 662	90	Capital

Introduction

Since this is a sample, we denote the average by $\bar{x}$:

$$\begin{aligned}\bar{x} &= \frac{\sum x_i}{n} = \frac{88 + 68 + 56 + 48 + 90}{5} \\ &= 70,\end{aligned}$$

but notice that the sample average $\bar{x} = 70$ is slightly different from the population average $\mu = 71$.

Introduction

We call $\bar{x}$ a **point estimate**. This is a statistic that is computed from sample data. We hope that it is a good enough estimate **of the population statistic** μ.

Now suppose we collected three additional random samples from the population of 800 schools in Bahrain, each containing only 5 schools.

The following tables present the average mathematics grades for each sample:

Introduction

Table 1: Three Different Samples of Schools in Bahrain

Sample 1

School	Grade x
12	65
25	72
48	68
76	74
91	71

$\bar{x}_1 = 70$

Sample 2

School	Grade x
103	78
156	82
204	70
318	75
422	80

$\bar{x}_2 = 77$

Sample 3

School	Grade x
115	60
230	65
345	68
470	62
590	70

$\bar{x}_3 = 65$

Introduction

This illustrates an important concept known as **sampling variability**.

That is, different random samples from the same population will generally produce different sample statistics.

Although each sample mean provides a **point estimate** of the population mean, some estimates will be closer to the true population mean than others.

Confidence Intervals for Continuous Variables

When statisticians generate data from a sample, how confident are they that the computed point estimates are actually equal to the true population parameters?

From our example earlier, we are essentially asking, "how sure are we that $\bar{x} = \mu$?".

Since $\bar{x}$ can produce different numbers based on the sample, it is like saying that $\bar{x}$ is *also* a random number, which is drawn from a distribution and which has a mean and variance.

Confidence Intervals for Continuous Variables

In order for us to analyse the level of confidence we have that our point estimates like $\bar{x}$ are accurate, let us start by invoking a powerful declaration called the **central limit theorem**.

Essentially, the **central limit theorem** says that if your sample is sufficiently large, then the **distribution** of **your point estimate** is **approximately normal**:

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

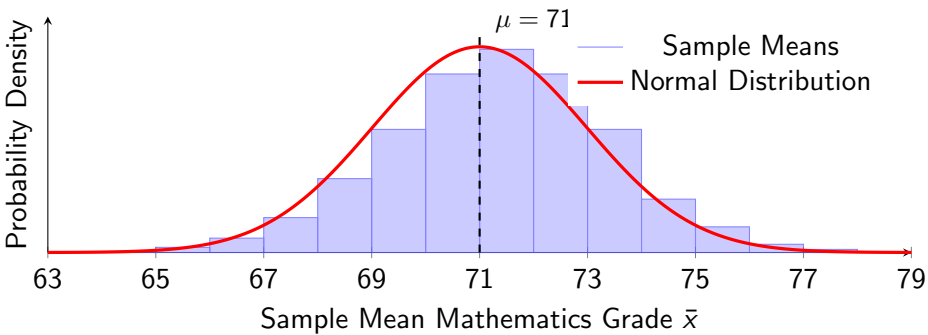
Confidence Intervals for Continuous Variables

which is parametrised by the true population mean μ and its variance $\frac{\sigma^2}{n}$. Its standard deviation is therefore the square root of its variance:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

Confidence Intervals for Continuous Variables

Figure 1: Illustrative Sampling Distribution of the Sample Mean ($n = 25$)



Confidence Intervals for Continuous Variables

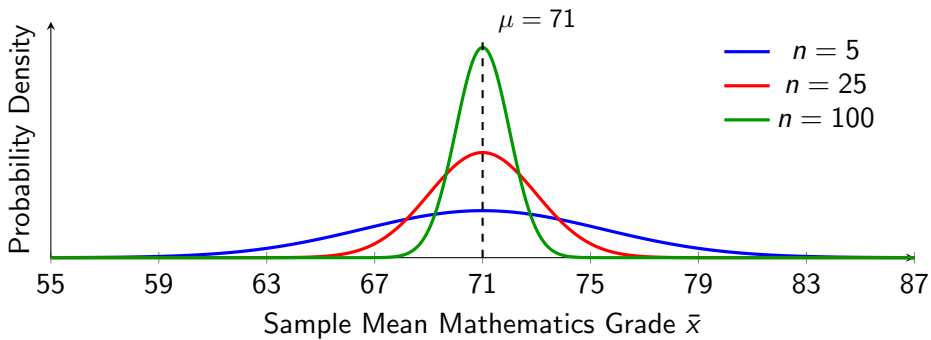
This images shows the $\bar{x}$ of different samples of $n = 25$ from the population.

Then the central limit theorem states that most estimates have a probability to be around the true mean μ, while the probability is much lower when it is away from μ.

Confidence Intervals for Continuous Variables

An interesting result is that the standard deviation has $\sqrt{n}$ in the denominator. As n gets very large, the distribution becomes very narrow:

Figure 2: Sampling Distribution of the Sample Mean for Different Sample Sizes



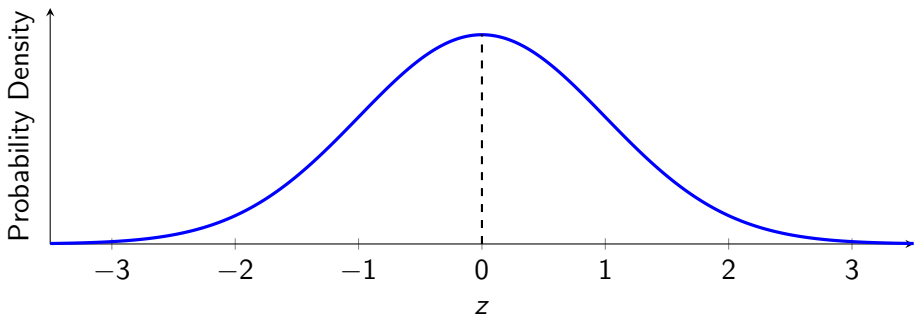
Confidence Intervals for Continuous Variables

Now just like we did in the last chapter, it does not actually matter what value is represented in the horizontal axis. So long as the shape is normally distributed, then we can standardise the values of $\bar{x}$:

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Confidence Intervals for Continuous Variables

Figure 3: Standard Normal Distribution



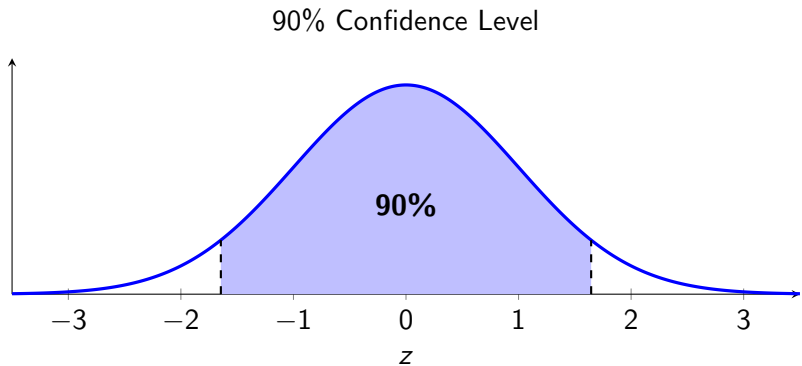
Confidence Intervals for Continuous Variables

Before we continue, it is important to remember this fact:

- 90% of the distribution area is between -1.64 and +1.64 standard deviatons.
- 95% of the disitribution area is between -1.96 and +1.96 standard deviations.
- 99% of the distribution area is between -2.58 and +2.58 standard deviations.

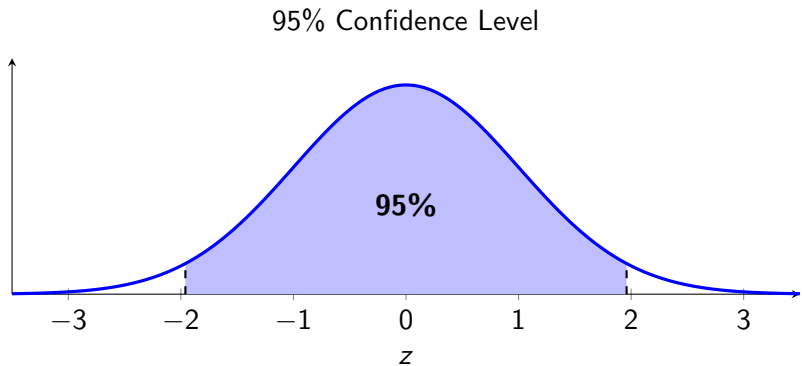
Confidence Intervals for Continuous Variables

Figure 4: Confidence Levels and the Standard Normal Distribution



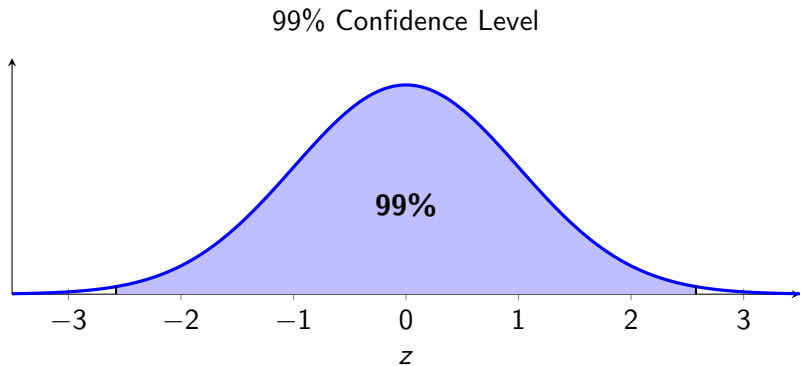
Confidence Intervals for Continuous Variables

Figure 5: Confidence Levels and the Standard Normal Distribution



Confidence Intervals for Continuous Variables

Figure 6: Confidence Levels and the Standard Normal Distribution



When σ is a Known Number

Now let us start with the idea of confidence intervals. This is essentially a measure for how confident we are that the computed point estimate $\bar{x}$ is within some interval that contains the true μ. The formula is given by:

$$\bar{x} \pm z\left(\sigma_{\bar{x}}\right)$$

$$\bar{x} \pm z\left(\frac{\sigma}{\sqrt{n}}\right)$$

where z describes the **level of confidence** that interval contains μ around the estimate $\bar{x}$.

When σ is a Known Number

The conventional levels for values of z is:

- 90% confident: z = 1.64.
- 95% confident: z = 1.96.
- 99% confident: z = 2.58.

When σ is a Known Number

Consider an example where a sample of $n = 25$ schools reveals that $\bar{x} = 68$ and that you are told that the population standard deviation is $\sigma = 9.95$.

Let us construct the 95% confidence intervals that the population mean is within those bounds:

When σ is a Known Number

$$= \bar{x} \pm z\left(\frac{\sigma}{\sqrt{n}}\right)$$

$$= 68 \pm 1.96\left(\frac{9.95}{\sqrt{25}}\right)$$

$$= 68 \pm 1.96(1.99)$$

$$= 68 \pm 3.9004$$

Note that the value 3.9004 is known as the **margin of error**.

When σ is a Known Number

Hence, the 95% confidence interval is:

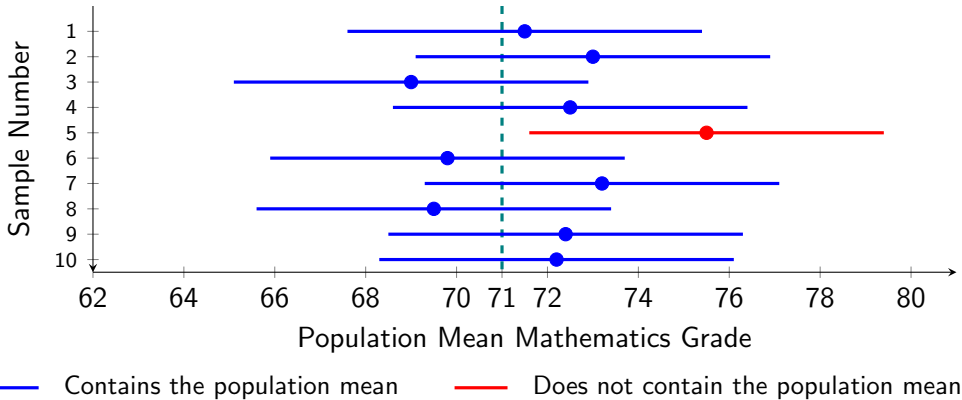
$$64.10 \leq \mu \leq 71.90$$

The way we interpret this that you are 95% confident that the true population mean mathematics grade of schools in Bahrain lies between 64.10 and 71.90.

This means that if we repeatedly collected random samples and constructed confidence intervals using this method, approximately 95% of those intervals would contain the true population mean μ.

When σ is a Known Number

Figure 7: Illustration of 20 Confidence Intervals at the 95% Confidence Level



When σ is Unknown

Remember that the population variance is given by:

$$\sigma^2 = \frac{\sum(x_i - \mu)}{N}$$

That meant that we could define the **normal** distribution of our estimator $\bar{x}$:

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

When σ is Unknown

However, suppose that σ^2 is unknown to the statistician. This means that we cannot compute the variance of $\bar{x}$.

So in this case, statisticians should instead use the **sample** variance:

$$s^2 = \frac{\sum(x_i - \bar{x})}{n-1}$$

When σ is Unknown

The variance and standard deviation of our $\bar{x}$ therefore become:

$$s_{\bar{x}}^2 = \frac{s^2}{n}$$

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

When σ is Unknown

However there is a consequence to doing this. If we replace the population parameter σ^2 with sample parameter s^2, then **we can no longer say** that $\bar{x}$ is normally distributed[a].

We instead have to say that $\bar{x}$ follows a **t-distribution**:

[a] A simple explanation of this is because $\bar{x}$ is **already** a sample estimate of the population. Now if we introduce the sample s^2 then this is also **another** estimate of the population. There is going to be variation in the values of $\bar{x}$ and s^2 for different samples. Therefore, each sample will have its **own unique** distribution shape, and therefore it is not a normal distribution

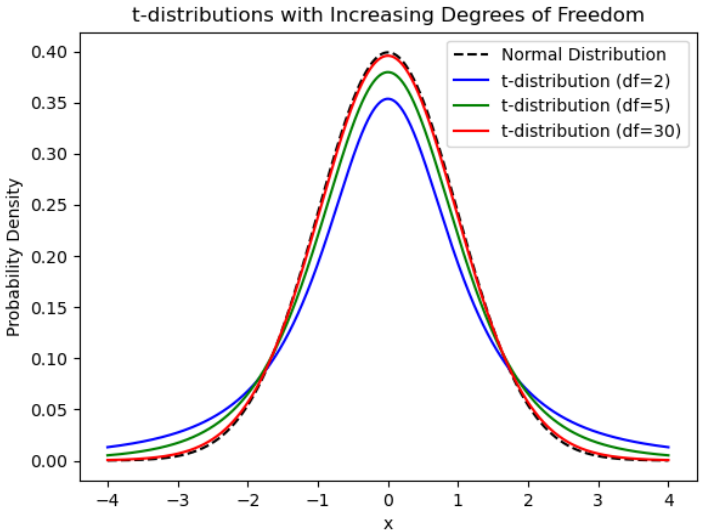
When σ is Unknown

$$T = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

where $n - 1$ is known as the **degrees of freedom**.

The t distribution actually looks like a normal distribution and has the same symmetric bell-shaped properties. It is only that the degree of how wide it becomes is based on the degrees of freedom:

When σ is Unknown



When σ is Unknown

Notice how when the degrees of freedom $n-1$ is very large, it essentially approximates the normal distribution (the red line and the dashed black line).

When σ is Unknown

In order to construct confidence intervals for $\bar{x}$ when σ is unknown, we need to use the below formula:

$$\bar{x} \pm t\left(\frac{s}{\sqrt{n}}\right)$$

Previously, we were using z values, such as 1.64, 1.96 or 2.58 depending the level on confidence. We used those values because the underlying distribution of $\bar{x}$ was normal.

However, because σ is unknown and we are using the sample equivalent s instead, then we are now in the t distribution and so we must use the t table instead:

When σ is Unknown

Table 2: Critical Values of t-Distribution

	Confidence Intervals				
df	**80%**	**90%**	**95%**	**98%**	**99%**
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
∞	1.282	1.645	1.960	2.326	2.576

When σ is Unknown

Suppose you have a sample of 10 schools and that the average score of the sample was $\bar{x} = 60$ and we do not have σ but the sample standard deviation $s = 8$. We can construct the 95% confidence intervals as follows:

$$\bar{x} \pm t\left(\frac{s}{\sqrt{n}}\right)$$

$$60 \pm t\left(\frac{8}{\sqrt{10}}\right)$$

When σ is Unknown

Since $n = 10$, the degrees of freedom is $n - 1 = 9$. For 95% confidence intervals, we select $t = 2.262$:

Table 3: Critical Values of t-Distribution

Confidence Intervals					
df	**80%**	**90%**	**95%**	**98%**	**99%**
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169

When σ is Unknown

$$60 \pm t\left(\frac{8}{\sqrt{10}}\right)$$

$$60 \pm 2.262\left(\frac{8}{\sqrt{10}}\right)$$

$$60 \pm 2.262(2.530)$$

$$60 \pm 5.72$$

When σ is Unknown

Hence, the 95% confidence interval is:

$$54.28 \leq \mu \leq 65.72$$

Confidence Intervals for Discrete Variables

We will now learn how to construct confidence intervals for discrete variables. Consider again the school data example:

School i	**Average Mathematics Grade**	**Pass Indicator X_i**
School 1	87	1
School 2	58	1
School 3	46	0
School 4	48	0
School 5	80	1
School 6	67	1
School 7	35	0
School 8	67	1
School 9	82	1
	⋮	
School 800	82	1

Confidence Intervals for Discrete Variables

Where the pass indicator is a random binary variable that takes values 0 and 1 depending on whether the grade is below or above 50 respectively.

Suppose we are interested in the average number of schools that passed. We can do this by simply summing up the individual columns:

$$\pi = \frac{\sum x_i}{n} = \frac{\sum 1+1+0+0+1+1+0+1+...+1}{800} = 0.65$$

where the average variable π represents the **proportion** of schools that passed. In this case, 65% of schools received a grade higher than 50.

Confidence Intervals for Discrete Variables

The proportion is basically the average but for discrete variables that can only take values between 0 and 1.

For example, a coin toss or a person's sex. However, you should note that in the previous example, we used **all** 800 schools to get the value of π. This means that $\pi = 0.65$ is the **population proportion**.

Confidence Intervals for Discrete Variables

Now let us examine the proportion when we have a sample:

Table 4: Three Different Samples of Schools in Bahrain

Sample 1

School	Pass x
421	0
352	1
284	1
267	1
719	1

$p_1 = 0.8$

Sample 2

School	Grade x
103	1
156	1
204	0
318	0
422	1

$p_2 = 0.6$

Sample 3

School	Grade x
115	1
230	0
345	0
470	0
590	1

$p_3 = 0.4$

Confidence Intervals for Discrete Variables

Just like when we looked at $\bar{x}$, the sample proportion p is also variable across samples. Moreover, we can similarly appeal to the central limit theorem and say that p has a (approximately) normal distribution, parametrised by its mean and variance:

$$p \sim N\left(p, \frac{p(1-p)}{n}\right)$$

Confidence Intervals for Discrete Variables

You are not expected to know how to derive the variance of p but you should know that the **standard deviation** is simply the square root of the variance:

$$\sigma_p = \sqrt{\frac{p(1-p)}{n}}$$

When σ is a Known Number

Now that we have the standard deviation, it is possible to construct confidence intervals for our sample proportion p:

$$p \pm z\left(\sigma_p\right)$$

$$p \pm z\left(\sqrt{\frac{p(1-p)}{n}}\right)$$

where just as before, we use the z values based on the significance level:

When σ is a Known Number

- 90% confident: $z = 1.64$.
- 95% confident: $z = 1.96$.
- 99% confident: $z = 2.58$.

However, you should keep in mind that we are **only able to do this** because we are appealing to the central limit theorem. That is, only if we have **a sufficiently large sample**, are we able to let the random variable x follow a normal distribution.

When σ is a Known Number

When you are dealing with a discrete number, you need to therefore keep this rule in mind:

When σ is a Known Number

You are only able to appeal to the central limit theorem if your sample satisfies:

- $np \geq 5$.
- $n(1-p) \geq 5$

When σ is a Known Number

For example, suppose $n = 10$ and $p = 0.5$. This means that:

- $np = 10 \cdot 5 = 5 \geq 5$.
- $n(1 - p) = 10(1 - 0.5) = 5 \geq 5$

then this qualifies as a large sample and so you can use the z values.

When σ is a Known Number

Activity:

Consider a sample dataset of $n = 2000$ people, of which 1600 are female. Let the binary variable x indicate $x = 0$ if male and $x = 1$ if female.

(a) Calculate the sample proportion p of females in the dataset.

(b) Verify if the sample satisfies the large sample conditions which allow you to appeal to the central limit theorem.

(c) Construct the 95% confidence interval.

When σ is a Known Number

Solution:

(a) The sample proportion:

$$p = \frac{1600}{2000} = 0.8$$

which shows that 80% of the individuals were women.

When σ is a Known Number

(b) The large sample conditions are:

- $np = 2000 \cdot 0.8 = 1600 \geq 5$
- $n(1-p) = 2000 \cdot (1-0.8) = 400 \geq 5$

When σ is a Known Number

(c) Remember that the formula is:

$$p \pm z\left(\sqrt{\frac{p(1-p)}{n}}\right)$$

- The sample proportion $\bar{p} = 0.8$.
- The 95% confidence interval implies $z = 1.96$.
- The sample size $n = 2000$.

When σ is a Known Number

Therefore:

$$0.8 \pm 1.96\left(\sqrt{\frac{0.8(1-0.8)}{2000}}\right)$$

Lower Limit of CI $= 0.782$ Upper Limit of CI $= 0.818$